Seventeen Features of a Data-Generating Process for Simulating Aggregated N-of-1 Trials: A Framework for Comparison

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Author. pmsimstats team

Revision 2026-10-07. Expanded from ten to seventeen features. Features 11 to 17 (Sections 13 to 19) were added after review. They meet the second and third selection criteria of Section 2, but not yet the first: their effects are derived or taken from the literature, not simulated in this compendium.

Review 2026-10-07. Checked against `analysis/report/NOTATION.md`, the package source and the stress-test output. Symbols that collided with canonical ones were renamed (expectancy $e_t \to \eta_t$; unexplained sensitivity $u_i \to v_i$; reliability, measurement error, unblinding, dropout and latent-scale symbols; see the notation index). Section 20 now records that the package anchors its carryover (it no longer compounds), and Section 8 reports the completed prognostic-baseline cells.

Purpose. A Monte Carlo study of an N-of-1 design can only show what its data-generating process (DGP) lets the data contain. This paper identifies seventeen features of a DGP that determine what a simulation of an aggregated N-of-1 trial can and cannot reveal. They can be used to describe, compare and choose among DGPs.

For each feature we give the intuition, the mathematics, the relevant literature, the choices made in this compendium and the evidence for them, and a best-practice recommendation. Section 20 compares five DGPs on all seventeen features:

- the published process of Hendrickson et al. (2020; "RH");
- the package's current construct;
- the refined process proposed in paper 01;
- mean moderation;
- the pharmacodynamic sensitivity model of `docs/38`.

Section 21 is a summary written for the introduction of paper 01. Evidence labels follow the compendium convention: derived, verified (computed or simulated), inspected, argued.

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Notation index and glossary

Notation follows `analysis/report/NOTATION.md`. Symbols marked *local* are defined here.

Table 1: Notation index: symbols, meanings and status

Symbol	Meaning	Status
Y_{it}	symptom score of participant i at visit t; $t = 0$ is baseline	canonical
BL_i	baseline level	canonical
BR_{it}, PB_{it}, TV_{it}	drug, expectancy (placebo) and natural-course responses	canonical
ε_{it}	observation-level residual (serial noise in the sensitivity model)	canonical
B_i, b_i	biomarker; standardized biomarker	canonical
D_{it}, D_{bc}	binary drug state; exposure-decayed drug indicator (code `Dbc`)	canonical
t_{od}, t_{sd}	time on drug in the current on-drug period; time since discontinuation	canonical
$t_{1/2}$	carryover (response) half-life	canonical
η_t	expectancy weight of visit t: 1 open label, 0.5 blinded, 0 at baseline (code `e`; `e_t` in `docs/40`, `docs/49`)	canonical (paper 06)
w_t	calendar week of visit t	local
$G(t_{od}; m, d, r)$	modified Gompertz curve with maximum m, displacement d, rate r; G_{TV} for the natural course	local
$\mu_{BR,t}$, $\mu_{BR,\mathrm{stop}}$	mean drug response at visit t; at discontinuation	local
$f(t)$, f_0, f_{res}	population drug-response time course, scaled to 1 at full effect; its starting and residual levels	local (`docs/38`)
$E(t)$	target (input) of the turnover model; E_{it} in Feature 13, the exposure	local
k_{out}, $t^R_{1/2}$	turnover rate constant; response half-life, $k_{\mathrm{out}} = \ln 2 / t^R_{1/2}$	local
g_t, $\mathbf{g}$, $\bar{g}_{\mathrm{off}}$	coupling profile (fraction of the biomarker's moderation present at visit t); its vector over the response block; its mean over off-drug visits	local (`docs/37`)
c_{bm}	covariance-moderation strength (correlation)	canonical
β_{bm}	mean-moderation strength (multiplier)	canonical

Symbol	Meaning	Status
σ_{BR}, σ_{bm}	SDs of the drug response and of the biomarker	canonical
σ_{TV}, σ_{BL}, σ	SDs of the natural course and the baseline; vector of component SDs	local
Σ_{BR}, M	covariance of the drug-response visits; correlation matrix of the response block	local
ω_i, $\bar{\omega}$, ζ, v_i, σ_v	participant drug sensitivity, its mean, the biomarker's effect on it, unexplained sensitivity and its SD	local (`docs/38`; v_i is `docs/44`'s u_i, renamed here because u_i is the canonical random intercept)
σ_ε	SD of the serial noise	local
f_B, f_{BR}	between-class variance fractions of the biomarker and the drug response in a latent responder-class mixture	local (report 03)
$c_{bm,TV}$, $c_{bm,PB}$, $c_{bm,BL}$, β_{bm}^{TV}	prognostic couplings of the biomarker to the natural course, placebo response and baseline; mean-moderation form for the natural course	local (`docs/49`)
s_0, s_t, s_{prog}	slope of Y on b at baseline, at post-baseline visit t; prognostic slope	local (`docs/48`)
$\hat{\beta}_B$, q	fitted shared biomarker slope; share of s_{prog} it absorbs (`docs/48`'s f)	local
v_c	vector of the biomarker's correlations with component c over visits	local (`docs/35`)
ρ	AR(1) serial correlation per week	canonical
K, A	component correlation; temporal correlation kernel	local (`docs/35`)
c_1, $c_\times$	cross-component correlation at the same visit; at different visits	local (`docs/35`)
$\beta_{bm:D}$	biomarker-by-treatment interaction, the estimand	canonical
μ_h, β_h, $z(\cdot)$	intercept and slope of the dropout logit; standardization	local
B_i^*, ξ_i, R_B	observed biomarker; its measurement error; reliability	local

Symbol	Meaning	Status
$h(b)$, Δ, b^*, β_{lin}	moderation function; threshold effect and threshold; linear projection	local
R_i, ω_R, ψ_0, ψ_g	responder indicator; responder sensitivity; logit intercept; gating slope	local; ψ_g canonical
ϕ, Φ	standard normal density and distribution function	standard
E_{it}, a_{it}, $\bar{a}$, $\text{ramp}(t_{od})$, T_{tit}	exposure; adherence and its mean; titration ramp; titration duration	local
χ, $P_i(t)$, $\bar{P}$, $\bar{p}$, s_{pb}	unblinding parameter; participant placebo response at full expectancy, its mean; intercept and biomarker slope of P_i	local
$t_{1/2,i}$, σ_h, z_i	person-level half-life; its log-scale SD; standard normal draw	local
$Y_{i,\text{scr}}$, $\varepsilon_{i,\text{scr}}$, σ^2_{btw}, σ^2_{wth}	screening score; its transient error; between- and within-person variances	local
τ, z, $C(z)$, $c_{B,\text{scr}}$	eligibility threshold; its standardized value; inverse Mills ratio; biomarker correlation with the screening score	local
L, U, ℓ_{it}	instrument floor and ceiling; latent linear predictor	local

Glossary.

- **DGP.** The probability model from which simulated trials are drawn.
- **Predictive / prognostic biomarker.** Associated with the size of the treatment effect, or with the outcome irrespective of treatment.
- **Coupling.** How strongly the biomarker is tied to the drug response at a given visit.
- **Ceiling.** The largest biomarker effect for which a covariance-based DGP has a valid joint distribution.

1. Summary

Table 2: The seventeen DGP features, the question each answers, and the recommended choice

#	Feature	Question it answers	Recommended choice
1	Outcome decomposition	What is the score made of?	Baseline minus drug, expectancy and natural-course responses, each with its own clock

#	Feature	Question it answers	Recommended choice
2	Onset of the drug response	How does benefit build on drug?	A saturating time course in time on drug, with clinically read landmarks; an effect-compartment or turnover model as the principled form
3	Offset and carryover	How does benefit decay off drug?	Pharmacodynamic half-life, anchored at discontinuation, zero before first exposure
4	Re-exposure	What happens when the drug restarts?	Drug-specific: restart from zero for re-titrated drugs, from the residual otherwise
5	Encoding of the predictive biomarker	How is the interaction placed in the model?	A moderation profile that fades with the drug effect, in the mean or the covariance; never a step
6	Prognostic pathways	Can the biomarker act without the drug?	Optional couplings to natural course, placebo response and baseline
7	Within-person dependence	How do deviations persist over visits and across components?	Separable AR(1) plus a person-level term, valid for every schedule
8	Between-person heterogeneity of response	Do participants differ in response beyond the biomarker?	Explicit unexplained sensitivity; drug-response variance that shrinks off drug
9	Observation process	How are the latent responses observed?	Phase-specific measurement error, the visit schedule as data, specified missingness mechanisms
10	Validity, estimand and calibration	Is the DGP coherent, and what does the simulation estimate?	Valid by construction (stop, never repair); an explicit estimand; parameters calibrated and reported
11	Biomarker measurement and distribution	Is the biomarker observed exactly, and how is it distributed?	A reliability parameter below 1; a realistic distribution where the encoding allows it
12	Functional form of moderation	Is the drug response linear in the biomarker?	Linear as the reference, with a threshold or responder-mixture scenario
13	Exposure versus assignment	Does the participant receive the drug as assigned?	Exposure separated from assignment: titration ramp and adherence below 1 as stress scenarios

#	Feature	Question it answers	Recommended choice
14	Integrity of blinding	Does expectancy depend on the actual drug state?	An unblinding parameter, zero in the reference and positive in a stress scenario
15	Between-person variation in carryover	Do participants share one carryover half-life?	Log-normal person-level half-lives as a stress scenario
16	Selection and regression to the mean	How do enrollment criteria shape the sample?	A noisy screening score with an eligibility threshold, kept separate from the baseline visit
17	Outcome scale	Can simulated scores leave the instrument's range?	Check the range; use a bounded model when out-of-range scores are not rare

Features 1 to 10 are the ones on which existing DGPs differ. Features 11 to 17 are shared gaps: all five DGPs compared in Section 20 make the same simplifying choice on each.

Three cross-cutting findings motivate the list:

1. **Defects in features 3 and 5 together produced the published power collapse under carryover** (`docs/37`).
2. **Feature 7 bounds what feature 5 can represent** (`docs/34` to `docs/36`).
3. **Feature 6 decides which analysis is valid** (`docs/48`, `docs/49`).

A DGP comparison that varies one feature while holding the others fixed is the only way to attribute a result to a modeling choice.

2. Framing: what a DGP must do

A simulation study defines aims, data-generating mechanisms, estimands, methods and performance measures (the ADEMP structure of Morris, White and Crowther 2019 [1]; see also Burton et al. 2006 [2]). For N-of-1 designs, the DGP must do three things.

- **Represent the clinical phenomena the design is meant to resolve:** within-person contrasts over time, carryover, expectancy, and between-person differences in response.
- **Encode a well-defined estimand,** here the biomarker-by-treatment interaction $\beta_{bm:D}$.

- **Be valid and verifiable.** It must be a proper probability model, with every parameter interpretable and every result reproducible.

Published simulation studies of N-of-1 designs make these choices implicitly and differently [31, 32, 33, 34], and the CENT reporting statement asks for several of them to be reported for real trials [35]. The first ten features below are the places where DGPs for these designs actually differ. We selected them by three criteria:

- the choice changes at least one simulation result materially (verified in this compendium);
- it has a pharmacological or clinical interpretation;
- it can be stated and compared across DGPs.

Features 11 to 17 meet the second and third criteria. For the first, the evidence is a derivation or the literature, not a simulation in this compendium; each section says which.

3. Feature 1: outcome decomposition

What it is. The symptom score is built from separately generated components, each with its own driver:

$$Y_{it} = \mathrm{BL}_i - (TV_{it} + PB_{it} + BR_{it}).$$

Table 3: Components of the symptom score and the clock that drives each

Component	Driver
TV (natural course)	time in the trial
PB (expectancy)	time in the trial, scaled by the expectancy η_t
BR (drug response)	time on drug

Intuition. An N-of-1 contrast compares a participant on drug with the same participant off drug. What makes that contrast informative, or misleading, are the processes that change over the same weeks: the condition improves on its own, belief in treatment produces benefit, and blinding changes that belief. A DGP that generates only "drug effect plus noise" cannot show how a design copes with these.

Mathematics. Each component is observed only through the sum. Identification therefore comes from design variation:

- expectancy varies between the open-label and blinded phases;
- drug state varies within the blinded phase;
- time varies throughout.

In the Hybrid design, the open-label visits are all on drug at $\eta_t = 1$ and every off-drug visit is blinded at $\eta_t = 0.5$. Expectancy and drug are therefore partly confounded, and only the blinded visits separate them cleanly (`docs/40`, Section 7; `docs/49`, Section 2).

Literature. Decomposing response into drug action and disease progression is standard in pharmacometrics [3, 4], and placebo response has its own longitudinal models [28]. That expectancy adds benefit when a treatment is known to be given is shown by the open-versus-hidden paradigm [29]. Its dependence on trial design (open label against placebo-controlled) is modeled by [30]. The three-component construction for N-of-1 designs is RH's [5].

This compendium. All DGPs use RH's decomposition and Gompertz parameters. The component decomposition paper (06) and `docs/31` examine whether the components can be recovered.

Best practice.

- Generate each component on its own clock.
- State which design contrasts identify which component.
- Report component-level parameters on the outcome scale.

4. Feature 2: onset of the drug response

What it is. The time course of benefit after the drug is started.

Intuition. Benefit builds over weeks, through titration and the accumulation of effect. Early on-drug visits therefore show a partial effect, and a design whose on-drug periods are short sees mostly partial effects.

Mathematics. RH use a modified Gompertz curve in time on drug,

$$G(t_{od}; m, d, r) = m \, \frac{\exp(-de^{-rt_{od}}) - e^{-d}}{1 - e^{-d}},$$

with $m = 10.99$, $d = 5$ and $r = 0.42$ per week. Its landmarks:

- the inflection is at $t = \ln d / r = 3.8$ weeks;
- half of the maximum is reached at 4.7 weeks;
- 90% is reached at 9.2 weeks (`docs/40`, Section 4).

The pharmacometric alternative is a turnover (effect-compartment) model,

$$\frac{df}{dt} = k_{\text{out}}(E(t) - f(t)), \qquad k_{\text{out}} = \frac{\ln 2}{t^R_{1/2}},$$

in which onset and offset share one rate. On drug from level f_0, $f(t) = 1 - (1 - f_0)\, 2^{-t/t^R_{1/2}}$, so onset is exponential rather than sigmoid. A sigmoid onset follows if the target $E(t)$ itself rises with titration.

Literature.

- **Effect-compartment models** describe the lag between concentration and effect [6].
- **Indirect-response (turnover) models** describe responses that build and decay through production and loss [7].
- **Emax and sigmoid-Emax models** relate exposure to effect [8].

This compendium. RH's Gompertz onset is kept in every current DGP. `docs/38` proposes the turnover form, which is not yet implemented.

Best practice.

- **Read the onset clinically** before simulating: how long until half of the effect is reached. Check it against the design's on-drug period lengths.
- **Prefer an onset that is coherent with the offset** (Feature 3).

5. Feature 3: offset and carryover

What it is. How the drug response decays after the drug is stopped.

Intuition. Two kinds of carryover exist, and they act on very different time scales.

- **Pharmacokinetic carryover** is drug still in the body. For prazosin it is about 2.3 hours of plasma half-life [9], gone within a day.
- **Pharmacodynamic carryover** is benefit that persists after the drug has gone, set by how fast the response turns over: days to weeks.

At weekly visits, only the pharmacodynamic kind matters, so the simulation's half-life is a response half-life.

Mathematics. Anchored decay from the level at discontinuation:

$$\mu_{BR,t} = \mu_{BR,\text{stop}}\; 2^{-t_{sd,t}/t_{1/2}}.$$

RH's recursion instead decays the previous visit's already-decayed mean by the total time since discontinuation. At off-drug visits one and two weeks after stopping, the second is then at $2^{-(1+2)/t_{1/2}}$ instead of $2^{-2/t_{1/2}}$, so the decay compounds (derived; `docs/37`, Appendix A). Before first exposure $t_{sd} = 0$, but there is no drug effect, so the response is zero. A decay function evaluated at $t_{sd} = 0$ returns 1 and must not be used there.

That error, made in the analysis-side indicator of manuscript 02, cost the exposure-weighted analysis 0.26 to 0.30 in power in the crossover design (verified, `docs/45`, Section 8).

Literature.

- **Carryover is the classical problem of crossover designs** [10, 11].
- **Behavioral carryover** that outlasts the pharmacological effect has been modeled explicitly [12].
- **Washout and analytical treatments** in N-of-1 designs [13].
- **Causal estimation under carryover** in aggregated N-of-1 trials, with a simulated DGP [27].

This compendium.

- **Decay forms:** exponential, with Weibull as a sensitivity option (manuscript 02, the stress test).
- **Half-life grid:** 0, 0.5 and 1 week.
- **Pharmacological range:** 0.1 to 0.2 week for prazosin (paper 01).

Best practice.

- **Specify carryover as a pharmacodynamic half-life,** with its justification.
- **Anchor the decay at discontinuation.**
- **Code pre-exposure visits as unexposed.**
- **Vary the decay form as a sensitivity analysis.**

6. Feature 4: re-exposure

What it is. The drug response when the drug is restarted after an off-drug period.

Intuition. Either the response rebuilds from zero, which is right if the drug is re-titrated, or it continues from whatever remains of the earlier effect, which is right if the residual carries forward.

Mathematics.

- **Restart from zero:** $\mu_{BR,t} = G(t_{od})$ with t_{od} reset. In the Hybrid crossover, a participant at near-full response at week 10 is credited with $G(4) = 4.28$ of 10.99 at week 16, 39% of the maximum.
- **Turnover model:** restart from the residual is automatic, $f(t) = 1 - (1 - f_{\text{res}})\, 2^{-t/t^R_{1/2}}$.
- **Combined rules:** anchored offset with a zero restart can step the mean down at re-exposure, whenever the residual exceeds the restarted curve.

Literature. Re-titration after an interruption is standard prescribing practice for prazosin, because of first-dose hypotension. Turnover models give restart from the residual by construction [7].

This compendium. RH, the package and the refined process all restart from zero. Restart from the residual is roadmap step 7 of `docs/44` and is not implemented.

Best practice.

- **Make the rule an explicit, drug-specific choice.**
- **For re-titrated drugs, restart from zero.**
- **Otherwise, restart from the residual,** preferably through a turnover model.

7. Feature 5: encoding of the predictive biomarker

What it is. How the biomarker-by-treatment interaction enters the model. There are three forms:

- **Covariance binding (RH):** a correlation $c_{bm} g_t$ between the biomarker and BR_{it}.
- **Mean binding:** a shift of $\beta_{bm} b_i \sigma_{BR}\, g_t$ in BR_{it}.
- **Sensitivity binding** (`docs/38`): $BR_{it} = \omega_i f(t) + \varepsilon_{it}$ with $\omega_i = \bar{\omega}(1 + \zeta b_i) + v_i$.

Intuition. A predictive biomarker tells you how much drug effect to expect. Wherever the drug effect is absent, the biomarker should therefore predict nothing. What matters most is not *where* the interaction is placed but its profile over visits, g_t, and above all whether that profile fades with the drug effect.

Mathematics.

(a) Conditional equivalence (derived). Under joint normality,

$$E(BR_{it} \mid B_i) = \mu_{BR,t} + c_{bm} g_t \sigma_{BR} b_i, \qquad \text{Cov}(BR_{i\cdot} \mid B_i) = \Sigma_{BR} - c^2_{bm} \sigma^2_{BR}\, g g^\top.$$

So covariance binding is mean binding with exposure profile g_t and $\beta_{bm} = c_{bm}$, and a rank-one smaller residual covariance.

(b) Power depends on the profile through the coupling gap (derived, `docs/37`). For the paired-difference statistic,

$$\beta_{bm:D} = -c_{bm}\frac{\sigma_{BR}}{\sigma_{bm}}(1 - \bar{g}_{\text{off}}),$$

where $\bar{g}_{\text{off}}$ is the mean off-drug coupling. Power falls with the gap $1 - \bar{g}_{\text{off}}$, and the required sample size scales roughly as $1/\text{gap}^2$.

Table 4: Coupling profiles, their off-drug coupling, and the coupling gap under carryover

Profile	Off-drug coupling	Gap under carryover
RH's step rule	1 wherever any drug effect remains	0: the interaction vanishes at any positive half-life
Graded coupling	$2^{-t_{sd}/t_{1/2}}$	falls smoothly with the half-life
Sensitivity model	$f(t)$, proportional to the response	follows the response

On RH's own code, graded coupling took power from 0.13 back to 0.64 at a half-life of 0.1 week (verified, `docs/37`, Appendix A).

(c) The ceiling (derived; `docs/34`, `docs/35`). Covariance binding requires

$$c_{bm}^2\,\mathbf{g}^\top M^{-1}\mathbf{g} < 1,$$

by the Schur complement, where M is the response correlation block and $\mathbf{g}$ the coupling pattern (g_t at the drug-response entries, zero elsewhere). Under RH's compound symmetry the ceiling is 0.256 without carryover, below RH's effect sizes of 0.3 and 0.6. RH's code repaired the infeasible matrices silently: 20 of 54 at those effect sizes. Mean and sensitivity binding have no ceiling: under sensitivity binding the correlation is an output,

$$\text{Cor}(B, BR_t) = \frac{\bar{\omega}\zeta f(t)}{\sqrt{(\bar{\omega}^2\zeta^2 + \sigma_v^2)f(t)^2 + \sigma_\varepsilon^2}}.$$

Literature.

- **Mean moderation** is the standard representation of treatment-effect heterogeneity by a covariate [14, 15].

- **Random participant-by-treatment variation** is the standard description of individual response in replicated crossovers [16, 17].
- **RH were, to our knowledge, the first** to parameterize the interaction as a covariance in the N-of-1 literature [5].
- **Report 03** shows that covariance binding matches the second moments of a latent responder-class mixture, with $c_{bm}^2 = f_B f_{BR}$, the product of the between-class variance fractions.

This compendium.

- **The refined process** uses graded coupling.
- **Paper 01** compares the three encodings.
- **The sensitivity model** is proposed and not yet implemented.

Best practice.

- **Choose the profile first, and make it fade with the drug effect.**
- **Choose the placement second.** Covariance binding is acceptable below its ceiling. Use mean or sensitivity binding when larger effects are needed.
- **Report the strength on two scales:** as the correlation or multiplier, and as the implied slope in outcome points per SD of biomarker.

8. Feature 6: prognostic pathways

What it is. Whether the biomarker can be associated with the outcome through routes that do not involve the drug.

Intuition. Blood pressure might predict who improves on placebo, or who improves anyway. A DGP in which the biomarker touches only the drug response cannot reveal analyses that mistake such an association for a drug interaction.

Mathematics. Each pathway implies a slope of Y on b at baseline (s_0) and afterwards (s_t):

Table 5: Baseline and post-baseline biomarker slopes implied by each prognostic pathway (per SD of biomarker; $\sigma_{TV} = 10$ and the PB SD is $10\eta_t$; derived, `docs/49`, Section 2)

Pathway	s_0	s_t
natural course, constant	0	$-c_{bm,TV}\sigma_{TV}$
natural course, growing	0	$-\beta_{bm}^{TV}\sigma_{TV}\, G_{TV}(t)/\max G_{TV}$
placebo response	0	$-c_{bm,PB}\, 10\, \eta_t$
baseline severity	$c_{bm,BL}\sigma_{BL}$	$c_{bm,BL}\sigma_{BL}$

An analysis that shares one biomarker slope between baseline and the off-drug visits fits $\hat{\beta}_B \approx q\, s_{\text{prog}}$, for some $0 < q < 1$ set by the weight of the baseline row. It then reports a spurious interaction of about $(1-q)\, s_{\text{prog}}$ (heuristic, `docs/48`, Section 2, where the share is written f).

At a prognostic coupling of 0.3, the null rejection rate was up to 0.10 for a random-intercept-plus-AR(1) analysis (S) and up to 0.42 for an unstructured covariance (A). Adding the expectancy terms (F1) restored it to 0.040 to 0.047 in all four prognostic cells (verified, `docs/49`; stress-test summary `decision-screen-ds-nosym-all14.csv`). In the prognostic-baseline cell (cell 40, null), where the constancy assumption holds, S rejected 0.050, F1 0.040, G1 0.042 and G2 0.046, while the published analysis R rejected 0.075 (verified, same file).

Under separability the joint validity condition generalizes to $c_{bm,BL}^2 + \sum_{cc'} [K^{-1}]_{cc'} v_c^\top A^{-1} v_{c'} < 1$ (derived and verified, `docs/35`, Section 4.8, where $c_{bm,BL}$ is written c_{bl}).

Literature.

- **The predictive/prognostic distinction** is central to the PATH statement [14, 15].
- **Treating baseline as a response** imposes a constancy constraint [18].

This compendium. Prognostic couplings to the natural course, the placebo response and the baseline level exist in the stress-test driver. The baseline-severity cells (40 and 41, null and alternative) were added to the stress test in a protocol revision and are complete (results above; `docs/46`, script 11). Only the placebo coupling is in the package (`c.bm.pb`).

Best practice. Include prognostic pathways as standard stress scenarios whenever analyses are compared. A DGP without them can confirm an analysis but never refute it.

9. Feature 7: within-person dependence and cross-component structure

What it is. How a participant's deviations persist over visits, and how the components co-vary.

Intuition. The structure can carry permanent traits (compound symmetry), fading memory (AR(1)), or both. Separability means one clock for all three components: the link between components fades at the same rate as each component's own memory (`docs/35`, "In brief").

Mathematics.

- **Separable form.** $M = K \otimes A$, with K the component correlation and $A_{ts} = \rho^{|w_t - w_s|}$. Its eigenvalues are products of those of K and A, so it is valid for every visit schedule.
- **The non-separable alternative** (the package's construct, with same-visit cross-correlation c_1 and $c_\times \rho^{|w_t - w_s|}$ otherwise) fails whenever the smallest eigenvalue of A is at most $(c_1 - c_\times)/(1 - c_\times)$, which happens on dense schedules.
- **RH's construct** is a sum of a person-level and an occasion-level separable term, each valid.
- **The ceiling factors.** Its schedule term is a sum of per-switch costs, for example $1/(1 - \rho^{2\,\mathrm{gap}})$ for off to on, with the gap between the two visits in weeks.
- **Implied outcome covariance.** With the person-level baseline, and component SDs $\sigma = (\sigma_{TV}, \sigma_{PB}, \sigma_{BR})^\top$ constant over visits,

  $$\mathrm{Cov}(Y_t, Y_s) = \sigma_{BL}^2 + (\sigma^\top K \sigma)\, \rho^{|w_t - w_s|},$$

  a random intercept plus AR(1). That is the structure assumed by the random-intercept-plus-`corCAR1` analysis. Because the PB SD scales with η_t, the match holds only up to that scaling (`docs/35`, Section 1, item 6).

Consequences.

- **Power.** At the same c_{bm}, AR(1) data give much less power than compound symmetry: 0.335 against 0.721 in Hybrid at $c_{bm} = 0.25$ without carryover (derived, closed form for the paired-difference statistic; `docs/37`, Section 5.8).
- **Analysis validity.** A compound-symmetry working model applied to AR(1) data rejected a true null 0.15 to 0.16 of the time in the crossover design (verified, simulated; `docs/45`, Section 4).

Literature.

- **Random effects plus serial correlation plus measurement error** as the general longitudinal model [19, 20].
- **AR(1) with unequally spaced visits** [21].
- **Kronecker (separable) covariances** for multivariate repeated measures [22].

This compendium.

- **RH:** compound symmetry, $\rho = 0.8$.
- **The package:** non-separable AR(1).
- **The refined process:** separable AR(1) at $\rho = 0.7$.

Best practice.

- **Specify the structure generatively** (person-level term plus serial process), so that validity holds by construction.
- **State ρ and its basis.**
- **Compare effect sizes across structures on the slope scale.**
- **Check the analysis model against the structure the DGP implies.**

10. Feature 8: between-person heterogeneity of response

What it is. How much participants differ in their drug response beyond what the biomarker explains, and how the drug-response variance behaves over time.

Intuition. Two participants with the same blood pressure need not respond equally. Under RH's construct, the drug-response component has its full SD of 8 even off drug, where its mean is zero. That is as if participants had a random drug response with no drug present, which adds noise to every on-off contrast.

Mathematics. Under the sensitivity model,

$$\mathrm{Var}(BR_t) = (\bar{\omega}^2\zeta^2 + \sigma_v^2)\, f(t)^2 + \sigma_\varepsilon^2,$$

which shrinks to the serial-noise variance as the drug effect washes out. The participant-by-treatment variance σ_v^2 is the quantity that replicated designs estimate [16, 17]. It bounds the attainable biomarker correlation by $\bar{\omega}\zeta/\sqrt{\bar{\omega}^2\zeta^2 + \sigma_v^2}$ (derived, `docs/38`, Section 5). A latent-class (bimodal) response is a different heterogeneity, with nonlinear conditional mean (report 03).

Literature. Individual response and its variance components in N-of-1 and replicated crossover designs [16, 17, 23].

This compendium. RH, the package and the refined process all use a constant drug-response variance. The sensitivity model makes it proportional to the response. Report 03 studies latent classes.

Best practice.

- **Separate explained from unexplained heterogeneity,** that is, ζ from σ_v.
- **Let the drug-response variance follow the drug effect.**
- **Treat the distribution of response as a stated assumption,** normal or mixture.

11. Feature 9: the observation process

What it is. How the latent responses become recorded data: measurement error, its dependence on the study phase, the visit schedule, the scale, and missingness.

Intuition.

- **Ratings can be noisier in some phases.** Open-label and blinded phases can differ in rating noise, as well as in mean.
- **Visits are not equally spaced.** For example, the Hybrid design measures at weeks 4, 8, 9, 10, 11, 12, 16 and 20.
- **Participants drop out,** sometimes because they are doing badly.

Mathematics.

- **Phase-specific measurement error** adds a variance $\sigma^2_{\varepsilon,\text{phase}}$ to each visit, which a homoscedastic working model omits.
- **Missingness mechanisms.** Under MCAR or MAR, a likelihood analysis with all observed data is valid; under MNAR it is not [24, 25].
- **A MAR hazard** depending on the previous score, $\text{logit}\,\Pr(\text{dropout at } t \mid \text{present at } t-1) = \mu_h + \beta_h\, z(Y_{t-1})$, can be calibrated to a target dropout rate per simulated trial (`docs/46`).

Literature. Missing-data theory [24, 25]. Informative dropout in N-of-1 designs is the subject of paper 09.

This compendium. RH's censoring patterns; the stress test's phase measurement error and its MCAR and MAR dropout; paper 09's informative dropout. Under 20% MCAR or MAR dropout, the random-intercept-plus-AR(1) analyses with CR2 standard errors stayed within the study's single-cell cap of 0.075 (at most 0.068, DS under MCAR; that value is about 2.6 MCSE above 0.05, so it is mildly liberal rather than nominal in the strict sense). The published analysis R rejected 0.079 under MCAR dropout and 0.071 with phase-specific measurement error. The unstructured model with AIC selection (DA) rejected 0.075 under MAR (verified, stress test cells 16, 23 and 24, `summary-full.csv`).

Best practice.

- **Treat the schedule as data.** Use calendar time in every correlation.
- **Include phase-specific error and at least MCAR and MAR dropout** as stress scenarios.
- **State the mechanism of any missingness simulated.**

12. Feature 10: validity, estimand and calibration

What it is. The properties that make a DGP trustworthy as the truth against which analyses are judged.

Intuition. A simulation tests analyses against a stated truth. If the truth silently changes (repaired matrices), is undefined (no estimand), or is unrealistic (uncalibrated parameters), the results cannot be interpreted.

Mathematics and practice.

- **Validity.** Either the DGP is generative (valid by construction), or every matrix is checked and infeasible cells stop the run. RH's silent repair changed correlations by up to 0.10 at $c_{bm} = 0.6$ (verified, `docs/36`).
- **Estimand.** State the target, here the on-drug moderation slope, and whether an analysis with a binary drug indicator estimates it or an attenuated contrast.
- **Calibration.** Choose parameters from data where possible, and report effect sizes on the clinical scale. For prazosin, published trials give the plasma half-life [9] and a biomarker association on the Clinician-Administered PTSD Scale [26]. The simulated scale differs, so the mapping must be explicit.
- **Verifiability:**
  - a faithful reference arm reproducing the published process;
  - one change at a time;
  - common random numbers across arms;
  - a Monte Carlo standard error for every reported rate;
  - a written ADEMP protocol stating the comparisons [1].

This compendium.

- **Reference arm:** a reproduction of RH's code at commit 58b32a9.
- **Paired designs** throughout.
- **Exploratory stress test:** docs/46, a working protocol whose revisions (Amendments 1 and 2) are disclosed.
- **Pending:** calibration of effect sizes to trial data.

Best practice.

- **Stop, never repair.**
- **Define the estimand before the DGP.**
- **Keep a reference arm.**
- **Report ADEMP** [1].

13. Feature 11: biomarker measurement and distribution

What it is. Whether the biomarker that enters the analysis is the biomarker that moderates the response, and what distribution it has.

Intuition. A blood pressure reading, a genotype score or an imaging measure is observed with error. The drug response follows the true value; the analysis sees the measured one. A noisy biomarker spreads participants who are alike in truth across the measured range, which flattens the estimated interaction.

Mathematics.

- **Attenuation.** Let the observed biomarker be $B_i^* = B_i + \xi_i$ with independent error and reliability $R_B = \mathrm{Var}(B)/\mathrm{Var}(B^*)$. For a normal biomarker, $E[B_i \mid B_i^*]$ is linear in B_i^* with slope R_B, so the interaction coefficient on B^* is attenuated by R_B [36].
- **Power.** With the biomarker standardized before analysis, the coefficient is $\beta_{bm:D}\sqrt{R_B}$ and its standard error is essentially unchanged when the interaction explains a small share of the residual variance. The noncentrality therefore falls by $\sqrt{R_B}$, and the sample size needed for a given power rises by about $1/R_B$ (derived). At $R_B = 0.8$ a trial needs about 25% more participants.
- **Covariance encoding.** Under covariance binding the correlation the analysis can see is $c_{bm}\sqrt{R_B}$.
- **Distribution.** Covariance binding draws B_i inside the multivariate normal, so the biomarker is necessarily normal. A binary or skewed biomarker (a genotype, a thresholded laboratory value) can be represented only under mean binding. Precision for the interaction is proportional to the biomarker's variance in the enrolled sample (see also Feature 16).

Literature. Random measurement error and regression dilution [36]. The prazosin motivating biomarker is pretreatment blood pressure [26], which is measured with error.

This compendium. Every DGP measures the biomarker once, at baseline, without error, from a normal distribution. The published power figures are therefore upper bounds with respect to biomarker reliability (argued).

Best practice.

- **Add a reliability parameter** R_B and report power at plausible values, not only at $R_B = 1$.
- **Draw the biomarker from a realistic distribution** when the encoding allows it.
- **State whether power is quoted for the true or the measured biomarker.**

14. Feature 12: functional form of moderation

What it is. The shape of the relation between the biomarker and the size of the drug response.

Intuition. A linear interaction says each unit of biomarker adds the same benefit. Biology often suggests otherwise: a drug may work only above a receptor or blood-pressure threshold, or participants may be responders or non-responders, with the biomarker shifting the odds of being a responder. When the effect changes sign across the biomarker range, the interaction is qualitative [37].

Mathematics. Write the moderation as a function $h(b_i)$, so the on-drug response is scaled by $1 + h(b_i)$.

- **Linear:** $h(b) = \beta_{bm} b$, the form used throughout.
- **Threshold:** $h(b) = \Delta\, \mathbb{1}(b > b^*)$.
- **Responder mixture:** $\omega_i = \omega_R R_i$ with $R_i \sim \text{Bernoulli}\{\Pr(R_i = 1 \mid b_i)\}$ and $\text{logit}\Pr(R_i = 1 \mid b_i) = \psi_0 + \psi_g b_i$, with ψ_g the gating slope of paper 03.

A linear working model estimates the linear projection $\beta_{\text{lin}} = \text{Cov}\{h(b), b\}/\text{Var}(b)$. For a threshold at the median of a standard normal biomarker ($b^* = 0$), $\beta_{\text{lin}} = \Delta\, \phi(0) \approx 0.40\,\Delta$, and the linear term captures $\phi(0)^2/(1/4) \approx 0.64$ of the moderation variance. The linear test then has about 80% of the noncentrality of a test that uses the true form (derived).

Literature. Qualitative and non-crossover interactions [37]; the PATH statement on modeling heterogeneous treatment effects [14].

This compendium. Every DGP compared here is linear in the biomarker, including the sensitivity model, where ω_i is linear in b_i. Paper 03 (latent class mixture application) works with responder classes in the analysis, not in the DGP.

Best practice.

- **Keep linear moderation as the reference.**
- **Add one threshold and one responder-mixture scenario** to show what the linear working model loses.
- **Report the moderation on the clinical scale** at several biomarker values, not only as a coefficient.

15. Feature 13: exposure versus assignment

What it is. The difference between the drug a participant is assigned and the drug they receive: dose titration and adherence.

Intuition. Prazosin is not given at full dose from the first day. In the largest veterans trial it was escalated in divided doses over five weeks to a daily maximum of 20 mg in men and 12 mg in women [38]. Participants also miss doses. In both cases the response is driven by exposure, while the design and the analysis clocks (t_{od}, t_{sd}) run on assignment.

Mathematics.

- **Exposure.** Let $E_{it} = a_{it}\,\mathrm{ramp}(t_{od})$, with adherence $a_{it} \in [0, 1]$ and a titration ramp $\mathrm{ramp}(t_{od}) = \min(1, t_{od}/T_{\mathrm{tit}})$, for example $T_{\mathrm{tit}} = 5$ weeks [38]. The drug response is driven by E_{it}, for example as the input of the turnover model of `docs/38`.
- **Attenuation.** If the response is linear in exposure and adherence is independent of the biomarker, the assigned-drug contrast and the interaction are both scaled by the mean adherence $\bar{a}$. Power falls accordingly, and the sample size rises by about $1/\bar{a}^2$ (derived). Variable adherence also adds between-person variance (Feature 8).
- **Dependence on the biomarker.** If adherence depends on the biomarker, for example through side effects, the interaction is biased, in a direction set by that dependence (argued).

Literature. Undetected nonadherence in randomized trials leads to underestimated efficacy and inflated outcome variability [39]. Titration schedule for prazosin in PTSD [38].

This compendium. All DGPs assume full adherence. Titration is not modeled separately; the Gompertz onset of Feature 2 may absorb part of it during the first on-drug period. Feature 4 (restart from zero on re-exposure) is the drug-specific answer for re-titration.

Best practice.

- **Separate exposure from assignment** in the DGP.
- **Simulate adherence below 1** as a stress scenario, and one scenario in which adherence depends on the biomarker.
- **Model titration explicitly** when its duration is comparable with a treatment period.

16. Feature 14: integrity of blinding

What it is. Whether participants and raters can tell the drug state during the blinded phases, so that expectancy follows the actual drug.

Intuition. The expectancy η_t is fixed by design: 1 in open label and 0.5 throughout the blinded phases. If a drug has a perceptible effect, blinded participants may guess when they are on it. Expectancy then rises on drug and falls

off drug, and part of the placebo response is mistaken for drug response. Prazosin lowered supine systolic blood pressure by 6.7 mm Hg relative to placebo at 10 weeks [38]; whether such an effect unblinds participants is unknown (argued).

Mathematics.

- **Unblinding parameter.** In blinded phases let $\eta_{it} = \frac{1}{2} + \chi\,(D_{it} - \frac{1}{2})$, with $\chi = 0$ for intact blinding and $\chi = 1$ for full unblinding.
- **Bias in the drug effect.** With $PB_{it} = \eta_{it} P_i(t)$, the blinded on-off contrast picks up $\chi P_i(t)$, which biases the drug main effect by $\chi \bar{P}$.
- **Bias in the interaction.** If the placebo response depends on the biomarker, $P_i = \bar{p} + s_{pb}\, b_i$ (a prognostic placebo pathway, Feature 6), the interaction picks up $\chi\, s_{pb}$ (derived). Unblinding thus turns a prognostic pathway into a spurious predictive one. An expectancy term coded from the design (`docs/49`) is constant within the blinded phase and cannot absorb it.

Literature. The blinding index and the assessment of blinding success [40].

This compendium. All DGPs fix η_t by design; $\chi = 0$ throughout.

Best practice.

- **Add an unblinding parameter** χ, zero in the reference scenario, and include $\chi > 0$ combined with a prognostic placebo pathway as a stress scenario.
- **Recommend that real trials report a blinding index** [40].

17. Feature 15: between-person variation in carryover

What it is. Whether participants share one carryover half-life.

Intuition. Pharmacokinetic and pharmacodynamic parameters vary between people, and clinical trial simulation in pharmacometrics usually represents that variation explicitly [41]. If carryover lasts longer in some participants than in others, a single half-life in the DGP understates how heterogeneous the off-drug visits are, and a single half-life in the analysis is partly wrong for everyone.

Mathematics.

- **Person-level half-lives.** Let $t_{1/2,i} = t_{1/2} \exp(\sigma_h z_i)$ with $z_i \sim N(0,1)$, and coupling $g_{it} = 2^{-t_{sd}/t_{1/2,i}}$ off drug.
- **Mean profile.** The population-mean profile $\bar{g}(t_{sd}) = E\{2^{-t_{sd}/t_{1/2,i}}\}$ is a mixture of exponentials. Its decay rate falls with time since discontinuation, so at long t_{sd} it is dominated by the longest half-lives and decays more slowly than the profile at the median half-life (derived).

- **Consequences.** An analysis with a common half-life in D_{bc} fits the mean profile poorly at both short and long t_{sd}. The person-level coupling gaps also vary, which adds variance. To first order, power under a binary drug indicator still follows the mean coupling gap of `docs/37` (argued).

Literature. Between-subject variability in clinical trial simulation [41]; prazosin's plasma half-life [9]. We found no estimate of between-person variation in the pharmacodynamic carryover of prazosin in PTSD.

This compendium. All DGPs, including the sensitivity model, use one half-life (or one turnover rate) for all participants.

Best practice.

- **Simulate log-normal person-level half-lives** as a stress scenario.
- **Check that the AIC-selected** D_{bc} remains well sized when the half-life varies between participants.

18. Feature 16: selection and regression to the mean

What it is. The enrollment mechanism: who enters the trial, and on what measured value.

Intuition. PTSD trials enroll people whose screening score exceeds a threshold. Part of a high screening score is transient, so scores fall afterward whatever the treatment. This regression to the mean can look like natural course or placebo response. Selection also narrows the distribution of anything correlated with the score.

Mathematics.

- **Regression to the mean.** Let the screening score be $Y_{i,\mathrm{scr}} = \mathrm{BL}_i + \varepsilon_{i,\mathrm{scr}}$ with between- and within-person variances σ^2_{btw} and σ^2_{wth}, and enroll when $Y_{i,\mathrm{scr}} > \tau$. The expected fall after selection is $\sigma^2_{\mathrm{wth}}/\sqrt{\sigma^2_{\mathrm{btw}} + \sigma^2_{\mathrm{wth}}}\; C(z)$, with $C(z) = \phi(z)/\{1 - \Phi(z)\}$ and z the standardized threshold [42].
- **Range restriction.** If the biomarker is prognostic for baseline severity (Feature 6), selection on the screening score reduces its variance in the enrolled sample to $1 - c^2_{B,\mathrm{scr}} C(z)\{C(z) - z\}$ (for a standardized biomarker), where $c_{B,\mathrm{scr}}$ is the biomarker's correlation with the screening score. Interaction precision is proportional to that variance, so power falls (derived).
- **Within-person contrasts.** The blinded on-off contrast is largely protected, because regression to the mean affects all post-screening visits alike. Contrasts that lean on the baseline visit, including analyses that share the biomarker effect with baseline (`docs/48`), are not.

Literature. Regression to the mean and how to address it in design and analysis [42].

This compendium. All DGPs draw the baseline from an unrestricted distribution; there is no screening step. The stress-test cells with a prognostic baseline (cells 40 and 41) have no selection.

Best practice.

- **Simulate a noisy screening score** with an eligibility threshold.
- **Keep the screening value separate from the baseline visit** of the analysis.

19. Feature 17: outcome scale

What it is. The range and distribution of the recorded score.

Intuition. Symptom scales are bounded sums of item ratings. A Gaussian DGP can produce scores below the floor, particularly for strong responders late in an on-drug period. Under positive moderation those are the high-biomarker participants who carry the interaction.

Mathematics.

- **Bounded models.** Either generate on a latent scale and map to the instrument, $Y_{it} = L + (U - L)\,\mathrm{logit}^{-1}(\ell_{it})$, or use a beta model, which keeps simulated scores within the instrument's range even when residuals are added [43].
- **Censoring at the bounds** (clipping) is simple but distorts the mean and variance near the floor.
- **Consequence.** Near the floor, responses are compressed. The interaction is attenuated and the residual variance becomes heteroscedastic, falling where it matters most (argued).

Literature. Beta regression for simulating a bounded clinical scale [43].

This compendium. All DGPs generate an unbounded Gaussian score. The fraction of simulated scores outside the instrument's range has not been computed.

Best practice.

- **Report the fraction of simulated scores outside the range.**
- **Use a bounded model** when that fraction is not negligible.

20. The five DGPs compared

Table 6: The five DGPs compared on all seventeen features

Feature	RH (58b32a9)	Package `covar`	Refined (paper 01)	Mean moderation	Sensitivity model (`docs/38`)
1 Decomposition	BL, TV, PB, BR	same	same	same	same
2 Onset	Gompertz in t_{od}	Gompertz	Gompertz	Gompertz	turnover $f(t)$
3 Offset	compounding recursion	anchored	anchored	anchored or compounding	anchored (turnover)
4 Re-exposure	restart at 0	restart at 0	restart at 0 (residual: proposed, not implemented)	restart at 0	from residual
5 Biomarker encoding	covariance, step profile	covariance, graded	covariance, graded	mean, on-drug only (or decayed)	mean through sensitivity, proportional to f
6 Prognostic pathways	none	placebo only	natural course, placebo, baseline (driver)	possible	possible
7 Dependence	CS, $\rho = 0.8$	non-separable AR(1)	separable AR(1), $\rho = 0.7$	as chosen	rank-one plus serial
8 Response heterogeneity	constant variance, full off drug	same	same	same	shrinks with the response
9 Observation	censoring sets	censoring sets	phase error, MCAR/MAR (driver)	as chosen	as chosen
10 Validity	silent repair	silent repair in sampler	stop on infeasible	always valid	valid by construction

Feature	RH (58b32a9)	Package `covar`	Refined (paper 01)	Mean moderation	Sensitivity model (`docs/38`)
11 Biomarker measurement	exact, normal	exact, normal	exact, normal	exact; any distribution possible	exact; any distribution possible
12 Form of moderation	linear	linear	linear	linear	linear in ω_i
13 Exposure	full adherence; no titration	same	same	same	same (exposure could enter the turnover input)
14 Blinding	η_t fixed by design	same	same	same	same
15 Carryover variation	one half-life	same	same	same	one turnover rate
16 Selection	none	none	none	none	none
17 Outcome scale	unbounded Gaussian	same	same	same	same

Driver: implemented in the simulation drivers, not yet in the package. The package column was checked against `buildSigma()` in `R/generateData.R` on 2026-10-07 (inspected): carryover on the mean is anchored at the last on-drug value (the `last_on` anchor; `docs/44`, step 3), and an earlier version of this table wrongly listed it as compounding.

The five DGPs agree on every row from 11 to 17. Those features do not discriminate among the existing processes; they mark limitations that all of them share.

21. Summary for the introduction of paper 01

Monte Carlo studies of aggregated N-of-1 trials can only show what their data-generating process lets the data contain. We identify seventeen features of such processes. The first ten are the features on which existing processes differ:

1. the decomposition of the score into baseline, natural course, expectancy and drug responses;

2. the onset of the drug response;
3. its offset and carryover;
4. the response on re-exposure;
5. the encoding of the predictive biomarker;
6. prognostic pathways for that biomarker;
7. the within-person dependence over visits and across components;
8. between-person heterogeneity of response;
9. the observation process, including measurement error and missingness;
10. the validity, estimand and calibration of the process.

The remaining seven are simplifications that every process we examined shares:

11. a biomarker measured without error;
12. moderation linear in the biomarker;
13. exposure equal to assignment;
14. intact blinding;
15. one carryover half-life for all participants;
16. no selection on a screening score;
17. an unbounded outcome scale.

Three findings motivate the list. First, the power collapse under small carryover reported by Hendrickson et al. is produced jointly by Features 3 and 5. Their biomarker coupling stays at full strength wherever any drug effect remains, so the simulated interaction vanishes at any positive half-life. A coupling that fades with the drug effect removes the collapse. Second, Feature 7 bounds Feature 5. Under compound symmetry the covariance encoding cannot represent biomarker correlations above 0.256 without carryover, and the published effect sizes relied on silently repaired matrices; a separable AR(1) structure is valid for every schedule and raises the bound (to 0.440 at $\rho = 0.8$, and to 0.474 across designs, 0.480 in Hybrid, at $\rho = 0.7$). Third, Feature 6 decides which analysis is valid. If the biomarker is also prognostic, analyses that share its effect with the baseline visit report spurious interactions. We use the first ten features to specify a refined process and to compare it with the published one, and treat the remaining seven as stated limitations and directions for stress testing.

22. Limitations

- **The features are drawn from this compendium's experience** with one disease area (PTSD) and one drug (prazosin). Other settings may add features, for example binary or count outcomes, or multiple biomarkers.
- **Some recommendations rest on derivations** and have not been simulated: the turnover onset, restart from the residual, and the sensitivity model.

- **Features 11 to 17 have not been simulated in this compendium.** Their stated effects are first-order derivations (attenuation by reliability, the linear projection of a threshold, scaling by adherence, the unblinding bias, range restriction) or arguments; the derivations assume linear models and normal distributions.
- **"Every process we examined"** (Section 21) refers to the five DGPs of Section 20, only one of which (RH) is published, not to a systematic review of the literature.
- **Parameter calibration to trial data is outstanding.**

23. References

Each reference was verified against PubMed (PMID given) or the publisher's page, on 2026-10-07. Exceptions are noted.

1. Morris TP, White IR, Crowther MJ. Using simulation studies to evaluate statistical methods. *Stat Med* 2019;38(11):2074-2102. doi:10.1002/sim.8086. PMID 30652356.
2. Burton A, Altman DG, Royston P, Holder RL. The design of simulation studies in medical statistics. *Stat Med* 2006;25(24):4279-4292. doi:10.1002/sim.2673. PMID 16947139.
3. Holford N. Clinical pharmacology = disease progression + drug action. *Br J Clin Pharmacol* 2015;79(1):18-27. doi:10.1111/bcp.12170. PMID 23713816.
4. Chan PL, Holford NHG. Drug treatment effects on disease progression. *Annu Rev Pharmacol Toxicol* 2001;41:625-659. doi:10.1146/annurev.pharmtox.41.1.625. PMID 11264471.
5. Hendrickson RC, Thomas RG, Schork NJ, Raskind MA. Optimizing aggregated N-of-1 trial designs for predictive biomarker validation: statistical methods and theoretical findings. *Front Digit Health* 2020;2:13. doi:10.3389/fdgth.2020.00013. PMID 34713026.
6. Sheiner LB, Stanski DR, Vozeh S, Miller RD, Ham J. Simultaneous modeling of pharmacokinetics and pharmacodynamics: application to d-tubocurarine. *Clin Pharmacol Ther* 1979;25(3):358-371. doi:10.1002/cpt1979253358. PMID 761446.
7. Dayneka NL, Garg V, Jusko WJ. Comparison of four basic models of indirect pharmacodynamic responses. *J Pharmacokinet Biopharm* 1993;21(4):457-478. doi:10.1007/BF01061691. PMID 8133465.
8. Holford NHG, Sheiner LB. Understanding the dose-effect relationship: clinical application of pharmacokinetic-pharmacodynamic models. *Clin Pharmacokinet* 1981;6(6):429-453. doi:10.2165/00003088-198106060-00002. PMID 7032803.
9. Hobbs DC, Twomey TM, Palmer RF. Pharmacokinetics of prazosin in man. *J Clin Pharmacol* 1978;18(8-9):402-406. doi:10.1002/j.1552-4604.1978.tb02456.x. PMID 690251.

10. Senn S. *Cross-over Trials in Clinical Research*, 2nd ed. Chichester: Wiley; 2002. (Publisher page; DOI not confirmed.)
11. Jones B, Kenward MG. *Design and Analysis of Cross-Over Trials*, 3rd ed. Boca Raton: Chapman and Hall/CRC; 2014. doi:10.1201/b17537.
12. Shi D, Ye T. Behavioral carry-over effect and power consideration in crossover trials. *Biometrics* 2024;80(2):ujae023. doi:10.1093/biomtc/ujae023. PMID 38563531.
13. Duan N, Kravitz RL, Schmid CH. Single-patient (n-of-1) trials: a pragmatic clinical decision methodology for patient-centered comparative effectiveness research. *J Clin Epidemiol* 2013;66(8 Suppl):S21-S28. doi:10.1016/j.jclinepi.2013.04.006. PMID 23849149.
14. Kent DM, Paulus JK, van Klaveren D, et al. The Predictive Approaches to Treatment effect Heterogeneity (PATH) statement. *Ann Intern Med* 2020;172(1):35-45. doi:10.7326/M18-3667. PMID 31711134.
15. Kent DM, Steyerberg E, van Klaveren D. Personalized evidence based medicine: predictive approaches to heterogeneous treatment effects. *BMJ* 2018;363:k4245. doi:10.1136/bmj.k4245. PMID 30530757.
16. Senn S. Mastering variation: variance components and personalised medicine. *Stat Med* 2016;35(7):966-977. doi:10.1002/sim.6739. PMID 26415869.
17. Araujo A, Julious S, Senn S. Understanding variation in sets of N-of-1 trials. *PLoS ONE* 2016;11(12):e0167167. doi:10.1371/journal.pone.0167167. PMID 27907056.
18. Liu GF, Lu K, Mogg R, Mallick M, Mehrotra DV. Should baseline be a covariate or dependent variable in analyses of change from baseline in clinical trials? *Stat Med* 2009;28(20):2509-2530. doi:10.1002/sim.3639. PMID 19610129.
19. Diggle PJ. An approach to the analysis of repeated measurements. *Biometrics* 1988;44(4):959-971. doi:10.2307/2531727 (confirmed via Crossref). PMID 3233259.
20. Verbeke G, Molenberghs G. *Linear Mixed Models for Longitudinal Data*. New York: Springer; 2000.
21. Jones RH, Boadi-Boateng F. Unequally spaced longitudinal data with AR(1) serial correlation. *Biometrics* 1991;47(1):161-175. doi:10.2307/2532504 (confirmed via Crossref). PMID 2049497.
22. Galecki AT. General class of covariance structures for two or more repeated factors in longitudinal data analysis. *Commun Stat Theory Methods* 1994;23(11):3105-3119. doi:10.1080/03610929408831436.
23. Senn S. Sample size considerations for n-of-1 trials. *Stat Methods Med Res* 2019;28(2):372-383. doi:10.1177/0962280217726801. PMID 28882093.
24. Little RJA, Rubin DB. *Statistical Analysis with Missing Data*, 3rd ed. Hoboken: Wiley; 2019. doi:10.1002/9781119482260.
25. Molenberghs G, Kenward MG. *Missing Data in Clinical Studies*. Chichester: Wiley; 2007. doi:10.1002/9780470510445.
26. Raskind MA, Millard SP, Petrie EC, et al. Higher pretreatment blood pressure is associated with greater posttraumatic stress disorder

symptom reduction in soldiers treated with prazosin. *Biol Psychiatry* 2016;80(10):736-742. doi:10.1016/j.biopsych.2016.03.2108. PMID 27320368.

27. Gartner T, Schneider J, Arnrich B, Konigorski S. Comparison of Bayesian networks, G-estimation and linear models to estimate causal treatment effects in aggregated N-of-1 trials with carry-over effects. *BMC Med Res Methodol* 2023;23:191. doi:10.1186/s12874-023-02012-5. PMID 37605171.
28. Gomeni R, Lavergne A, Merlo-Pich E. Modelling placebo response in depression trials using a longitudinal model with informative dropout. *Eur J Pharm Sci* 2009;36(1):4-10. doi:10.1016/j.ejps.2008.10.025. PMID 19041717.
29. Colloca L, Lopiano L, Lanotte M, Benedetti F. Overt versus covert treatment for pain, anxiety, and Parkinson's disease. *Lancet Neurol* 2004;3(11):679-684. doi:10.1016/S1474-4422(04)00908-1. PMID 15488461.
30. Rutherford BR, Roose SP. A model of placebo response in antidepressant clinical trials. *Am J Psychiatry* 2013;170(7):723-733. doi:10.1176/appi.ajp.2012.12040474. PMID 23318413.
31. Wang Y, Schork NJ. Power and design issues in crossover-based N-of-1 clinical trials with fixed data collection periods. *Healthcare (Basel)* 2019;7(3):84. doi:10.3390/healthcare7030084. PMID 31269712.
32. Blackston JW, Chapple AG, McGree JM, McDonald S, Nikles J. Comparison of aggregated N-of-1 trials with parallel and crossover randomized controlled trials using simulation studies. *Healthcare (Basel)* 2019;7(4):137. doi:10.3390/healthcare7040137. PMID 31698799.
33. Tang J, Landes RD. Some t-tests for N-of-1 trials with serial correlation. *PLoS ONE* 2020;15(2):e0228077. doi:10.1371/journal.pone.0228077. PMID 32017772.
34. Chen X, Chen P. A comparison of four methods for the analysis of N-of-1 trials. *PLoS ONE* 2014;9(2):e87752. doi:10.1371/journal.pone.0087752. PMID 24503561.
35. Vohra S, Shamseer L, Sampson M, et al. CONSORT extension for reporting N-of-1 trials (CENT) 2015 Statement. *BMJ* 2015;350:h1738. doi:10.1136/bmj.h1738. PMID 25976398.
36. Hutcheon JA, Chiolero A, Hanley JA. Random measurement error and regression dilution bias. *BMJ* 2010;340:c2289. doi:10.1136/bmj.c2289. PMID 20573762.
37. Gail M, Simon R. Testing for qualitative interactions between treatment effects and patient subsets. *Biometrics* 1985;41(2):361-372. PMID 4027319. (JSTOR stable 2530862; DOI not confirmed.)
38. Raskind MA, Peskind ER, Chow B, et al. Trial of prazosin for post-traumatic stress disorder in military veterans. *N Engl J Med* 2018;378(6):507-517. doi:10.1056/NEJMoa1507598. PMID 29414272.
39. Le Flohic E, Vrijens B, Hiligsmann M. The impacts of undetected nonadherence in phase II, III and post-marketing clinical trials: an overview. *Br J Clin Pharmacol* 2024;90(8):1984-2003. doi:10.1111/bcp.16089. PMID

38752447.
40. Bang H, Ni L, Davis CE. Assessment of blinding in clinical trials. *Control Clin Trials* 2004;25(2):143-156. doi:10.1016/j.cct.2003.10.016. PMID 15020033.
41. Holford N, Ma SC, Ploeger BA. Clinical trial simulation: a review. *Clin Pharmacol Ther* 2010;88(2):166-182. doi:10.1038/clpt.2010.114. PMID 20613720. (Cited for general practice; its treatment of between-subject variability was not checked against the full text.)
42. Barnett AG, van der Pols JC, Dobson AJ. Regression to the mean: what it is and how to deal with it. *Int J Epidemiol* 2005;34(1):215-220. doi:10.1093/ije/dyh299. PMID 15333621.
43. Rogers JA, Polhamus D, Gillespie WR, et al. Combining patient-level and summary-level data for Alzheimer's disease modeling and simulation: a beta regression meta-analysis. *J Pharmacokinet Pharmacodyn* 2012;39(5):479-498. doi:10.1007/s10928-012-9263-3. PMID 22821139.